Social Choice Project

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1 Paper Summary

1.1 Paper title and authors, where was it published?

The paper *Preference Elicitation For General Random Utility Models* by Hossein Azari Soufiani, David C. Parkes, and Lirong Xia (SEAS, Harvard University) was published in AUAI 2013 (Association for Uncertainty in Artificial Intelligence).

1.2 What is the topic of the paper? What is the main claim? How does this relate to the material we studied in the course?

The paper focuses on *Preference Elicitation* within the framework of *General Random Utility Models* (GRUMs). The main claim of the paper is that its proposed adaptive elicitation scheme increases the precision of preference estimation in GRUMs.

This paper relates to the topic of *Truth Discovery*, covered in later parts of the course. Similar to the *Mallows Model* learned in class, this paper's model assumes that agents' observed preferences are noisy signals generated from an underlying ground truth. However, both models differ in the parametric assumptions they make and the form of their ground truth. The Mallows Model assumes a **single** global ground truth preference, and the noise is **distance-based** from the ground truth (higher noise means the observed preference is more distant from the ground truth). The GRUM model is **utility-based**, and assumes that utilities are influenced by the features of both agents and alternatives, meaning that each pair of agent and item has a unique ground truth utility, with the addition of an independent noise term.

1.3 What are the main 1-2 theoretical results in the paper? Why do you see them as the main results? Write them down as formally as possible, with all the required definitions

Definition 1 (General Random Utility Model - GRUM)**.** Let $\mathcal{C} = \{c_1, \ldots, c_m\}$ be a set of alternatives and a set of agents indexed by $i \in \{1, \ldots, n\}$. For every agent i and alternative j, the utility u_{ij} is defined as:

$$u_{ij} = \delta_j + \vec{x}_i B \vec{z}_j^\top + \epsilon_{ij}$$

Where:

- $x_i \in \mathbb{R}^K$ and $z_j \in \mathbb{R}^L$ are observed attribute vectors for agent i and alternative j (correspondingly).
- $\Theta = (\delta, B)$ are the parameters to be estimated.
  - $\delta \in \mathbb{R}^m$ is the vector of intrinsic utilities.
  - $B \in \mathbb{R}^{K \times L}$ is the vector intersection matrix.
- ϵ_{ij} is a random noise term.
- Denote the deterministic term as $\mu_{ij} = \delta_j + \vec{x}_i B \vec{z}_j^\top$.

Definition 2 (Preference Data)**.** Let $D = \{\pi_1, \ldots, \pi_n\}$ be the set of observed rankings, where π_i represents the full ranking of agent i such that $c_{\pi_i(1)} \succ_i \cdots \succ_i c_{\pi_i(m)}$.

Definition 3 (Identifiability)**.** The linear model describing the deterministic utility, $\mu_{ij} = \delta_j + \vec{x}_i B \vec{z}_j^\top$, in terms of parameters Θ is *identifiable* if the mapping from parameters to utilities is injective (one-to-one). That is, distinct parameters yield distinct deterministic utilities.

Definition 4 (Location Family Assumption)**.** The noise terms ϵ_{ij} are drawn from a distribution where the probability density function is log-concave (e.g., Gumbel or Normal).

Theorem 1 (Global Optimality)**.** *For the location family, if for every $i \leq n$ the joint probability density function for $\vec{\epsilon}_i$ and the prior $\Pr(\Theta)$ are log-concave, then $\Pr(\Theta \mid D)$ is concave up to a known transformation.*

Note. By *up to a known transformation*, the authors meant that the posterior is log-concave (the "known transformation" is the natural logarithm function). Also, the original Theorem 1 from the paper had a typo; it should have been *every* $i \leq n$ instead of *every* $j \leq m$ because we require that each agent's noise vector is drawn from a log-concave distribution.

Remark. Standard Expectation-Maximization (EM) algorithms suffer from the problem of getting stuck in local minima. Theorem 1 guarantees that the posterior is unimodal, ensuring that gradient-based inference methods will converge to the global maximum. In short, the theorem ensures the answer is **correct**.

Condition 1 (Connectivity)**.** Given the data D, in every partition of the alternatives C into two non-empty subsets $C_1 \cup C_2$, there exists $c_1 \in C_1$ and $c_2 \in C_2$ such that there is at least one ranking in D where $c_1 \succ c_2$.

Theorem 2 (Boundedness of Solutions)**.** *Suppose we fix $\mu_{11} = 0$. Then the set S_D of global maxima solutions to $\Pr(\Theta \mid D)$ is bounded in Θ if and only if the data D satisfies Condition 1 and the linear model describing μ in terms of Θ is identifiable.*

Remark. Theorem 2 provides the necessary and sufficient conditions to ensure the solution set is bounded (see Fig. 1). It generalizes Ford's condition for Plackett-Luce (Ford Jr 1957) to the broader class of GRUMs. In short, the theorem ensures the answer is **computable**.

1.4 Explain the outline of the proof in 2-3 paragraphs.

The original paper omitted the proofs for the theorem, and I was unable to find a version of the paper with the appendix of the proofs. After contacting the authors, they referred me to the Phd thesis of the main author (Soufiani 2014). Theorems 1 and 2 in the thesis can be generalized to prove the above theorems (correspondingly).

Proof of Theorem 1. Theorem 1 is proved by applying the following lemma:

Lemma 1. *Let $R \in \mathbb{R}$. Suppose $g_1(\vec{\Theta}, \vec{\epsilon}), \ldots, g_R(\vec{\Theta}, \vec{\epsilon})$ are concave functions in $\mathbb{R}^{m(2n+1)}$ where $\vec{\Theta}$ is the vector of $m + mn$ parameters and $\vec{\epsilon}$ is a vector of mn real numbers that are generated according to a distribution whose pdf is logarithmic concave in $\mathbb{R}^{mn}$. Then the following function is log-concave in $\mathbb{R}^m$:*

$$L_i(\vec{\Theta}, G) = \Pr\left[g_1(\vec{\Theta}, \vec{\epsilon}) \geq 0, \ldots, g_R(\vec{\Theta}, \vec{\epsilon}) \geq 0\right], \quad \vec{\Theta} \in \mathbb{R}^{m(n+1)}$$

A ranking π_i implies that the item at rank r, denoted $j = \pi_i(r)$, is preferred to the item at rank $r + 1$denoted $k = \pi_i(r + 1)$. This means the utility of the higher-ranked item is greater:

$$U_{\pi_i(r)} \geq U_{\pi_i(r+1)}$$

We can substitute this with the GRUM definition:

$$\delta_j + x_i B z_j^\top + \epsilon_{ij} \geq \delta_k + x_i B z_k^\top + \epsilon_{ik}$$

Lemma 1 requires the problem to be framed as probability constraints in the format $g(\Theta, \epsilon) \geq 0$. We rearrange the inequality above and define:

$$\begin{aligned} g_r^i(\Theta, \epsilon) &= U_{\pi_i(r)} - U_{\pi_i(r+1)} \\ &= \delta_j + x_i B z_j^\top + \epsilon_{ij} - \delta_k - x_i B z_k^\top - \epsilon_{ik} \\ &= (\delta_j - \delta_k) + x_i B (z_j - z_k)^\top + (\epsilon_{ij} - \epsilon_{ik}) \geq 0 \end{aligned}$$

The function g_r^i represents the pairwise comparison between rank r and $r+1$. Lemma 1 applies only if the functions g are **concave**.

- It's linear in δ because its a simple difference.
- It's linear in B because x, z are observed data (constants), so it's the same as multiplying by some constant C.

Since all linear functions are concave, the condition is satisfied.

1. The noise ϵ is assumed to have a **log-concave PDF** (e.g., Normal or Gumbel).
2. The constraints g are concave.
3. Therefore, by Lemma 1, the probability of observing a single ranking $\Pr(\pi_i|\Theta)$ is log-concave.
4. Since the sum of concave is also concave, the total likelihood, $\Pr(D \mid \Theta)$, is log-concave.

We also assumed that the prior, $\Pr(\Theta)$, is log-concave. This implies that the posterior, $\Pr(\Theta \mid D) \propto \Pr(D \mid \Theta) \cdot \Pr(\Theta)$, is log-concave. □

Proof of Theorem 2. Theorem 2's proof uses the following lemma:

Lemma 2. *If alternative j is preferred to alternative j' in at least one ranking then the difference of their mean parameters $\theta_{j'} - \theta_j$ is bounded from above ($\exists Q : \theta_{j'} - \theta_j < Q$) for all the $\vec{\theta}$ that maximize the likelihood function.*

If distinct sets of parameters Θ and Θ' yield the exact same utilities μ, the model cannot distinguish between them, and the solution set might be unbounded (e.g., along a line). Additionally, if Condition 1 does not hold, then S_D is unbounded because the parameters for all alternatives in C_1 can be increased simultaneously to improve the log-likelihood. Consider a directed graph G_D, where the nodes are the alternatives, and there is edge from c_j to $c_{j'}$ if in a least one ranking $c_j \succ c_{j'}$. By Condition 1, for any pair $j \neq j'$, there is a path from c_j to $c_{j'}$ (and conversely, a path from $c_{j'}$ to c_j). To see this, consider building a path between j and j' by starting from a partition with $C_1 = \{j\}$ and following an edge from j to j_1 in the graph where j_1 is an alternative in C_2 for which there must be such an edge, by Condition 1. Consider the partition with $C_1 = \{j, j_1\}$, and repeat until an edge can be followed to vertex $j' \in C_2$. It follows from Lemma 2 that for any $\vec{\theta} \in S_D$ we have $|\theta_j - \theta_{j'}| < Qm$, using the telescopic sum of bounded values of the difference of mean parameters along the edges of the path, since the length of the path is no more than m (and tracing the path from j to j' and j' to j), meaning that S_D is bounded. □

1.5 Create a figure/table that is not included in the paper, which can help clarify some of the definitions/results/proofs.

To clarify the implications of Theorem 2, I generated Figure 1, which visualizes the relationship between data connectivity (Condition 1) and the stability of the solution. GRUM estimates parameters (Θ) based on the difference in utilities between alternatives. If the data violates Condition 1 (as shown in the left graph of Figure 1), the alternatives are partitioned into disconnected sets, for example, $\{A, B\}$ never compete against $\{C\}$. Because the model relies on relative comparisons, there is no data to 'anchor' the utility of set $\{C\}$ against the others. The contour plot on the bottom left illustrates the consequence of this violation: the log-likelihood function forms an 'infinite ridge' rather than a peak. The white arrow demonstrates that we can increase the parameter θ_C indefinitely (shifting it to $+\infty$ or $-\infty$) without changing the likelihood of the

observed data. This shows that the solution set S_D is unbounded, meaning the algorithm cannot converge to a unique answer. In contrast, when the data satisfies Condition 1 (right graph), the comparison graph is strongly connected. This connectivity constrains the parameters relative to each other. As seen in the bottom right contour plot, this collapses the ridge into a strictly concave function with a single, bounded global maximum (the red dot), guaranteeing that the MAP estimate is unique and computable.

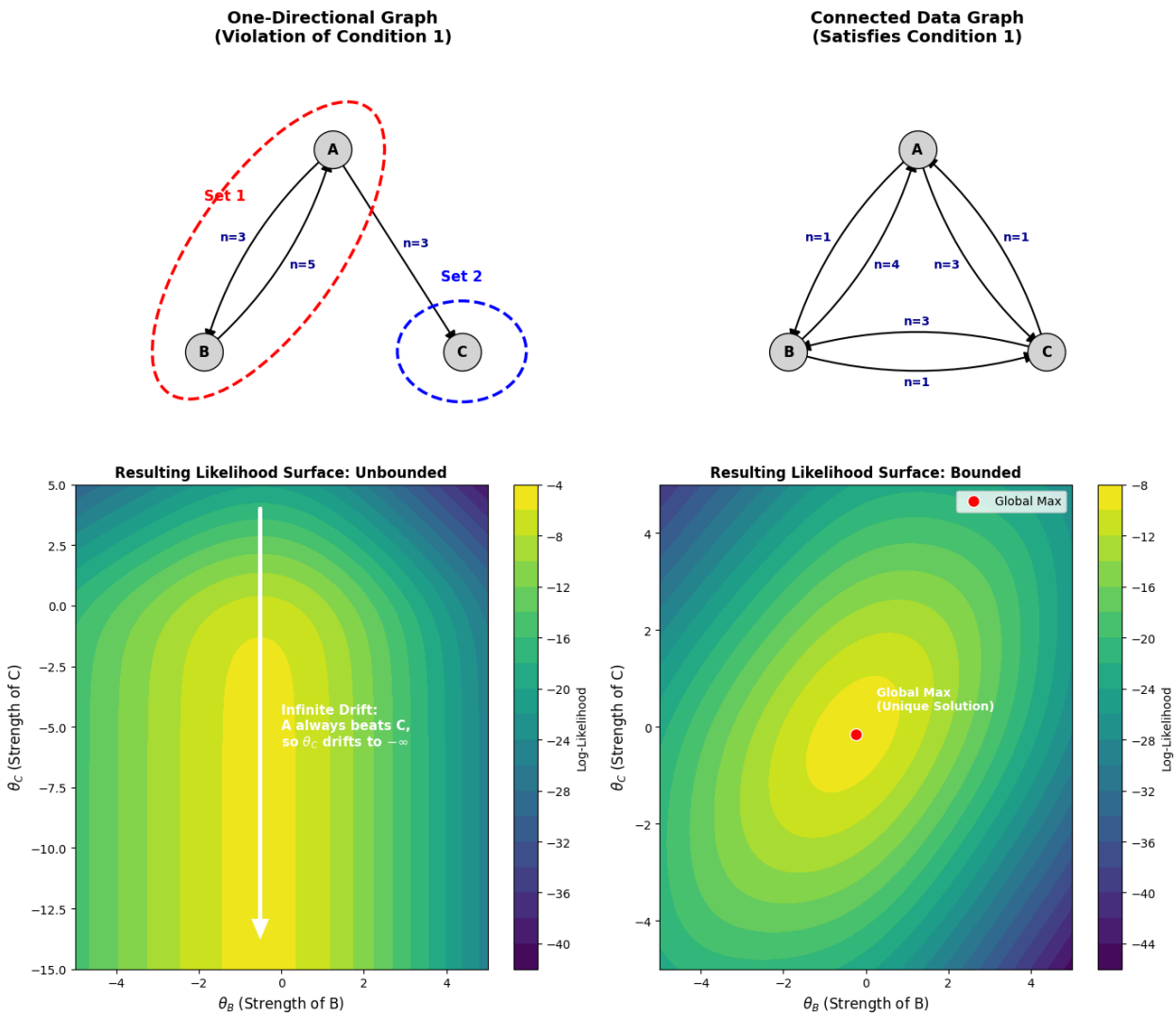

Figure 1: **Impact of Connectivity on Parameter Estimation.** Comparison of likelihood landscapes for disconnected vs. connected data. **(Left)** Data violating Condition 1 leads to an "infinite drift" in the likelihood surface, causing unbounded solutions where parameters cannot be uniquely identified. **(Right)** Data satisfying Condition 1 (strongly connected graph) ensures a unique, bounded global maximum for the GRUM parameters

Note that n represents "number of wins" in the pairwise comparisons we query; edges may not sum to the same number because we sample the comparisons and not measure all possible pairs.

1.6 Select one paper cited in your paper as an important paper, and explain (1-2 sentences) why it is cited and why you picked this particular paper.

The paper *Automobile Prices in Market Equilibrium* (Berry et al. 1993) (often referred to as the BLP model) was cited as an important paper because the authors explicitly define GRUMs as a specific case of this

influential generative model from econometrics. This cite establishes the theoretical foundation of the work, while distinguishing GRUMs as a simplified version that relies only on observed attributes to ensure the computational tractability that the original BLP model lacked.

1.7 Find (use Google Scholar) one paper citing your paper, and explain why it is cited.

One paper that cites this work is "*Knowledge Elicitation via Sequential Probabilistic Inference for High-Dimensional Prediction*" (Daee et al. 2017).

It cites this paper as a key example of preference model elicitation, referencing it to contextualize their own method. Specifically, they highlight it as a technique where expert knowledge (in the form of preferences) is sequentially queried to refine model parameters, similar to how they propose eliciting feedback to improve high-dimensional regression models.

1.8 Find another paper that should have been cited in your opinion, but isn't. Add a sentence to the "related work" part of the paper with the new citation.

The paper: *Preference learning with Gaussian processes* (Chu and Ghahramani 2005), presented in the International Conference of Machine Learning (ICML), 2005.

Reason: This paper is a seminal work in probabilistic preference learning. It establishes the foundational Bayesian framework for ranking with kernels. Citing it would contrast the authors' parametric approach (GRUMs) with the leading non-parametric Bayesian approach (Gaussian Process Preference Learning).

New sentence for related work: While non-parametric Bayesian methods such as Gaussian Processes have been successfully applied to preference learning (Chu and Ghahramani 2005), our GRUM-based approach offers a parametric alternative that explicitly models the structural relationship between agent and alternative attributes.

2 Project Plan

As part of my thesis research, I am investigating preference elicitation of Large Language Models (LLMs). This is done under the lens of economics, measuring the effect of such preferences on markets. This project applies GRUMs to LLM preference elicitation. Unlike the Bradley-Terry (BT) model, currently used in my thesis research, which treats prompt variations as random noise, we assume these variations create specific "personas" with distinct preferences. We will use GRUMs to model how the LLM's internal state (the "agent") interacts with the items (the "alternatives").

2.1 Methodology

We propose a framework where LLM prompts serve as agents (personas) and predefined token outputs (e.g., colors) as alternatives.

- **Agents (x_i):** We define the "agent" as the model's internal state after processing the initial prompt. We will use the **last hidden layer** to represent this state. Since these vectors are high-dimensional, we will project them down to a lower dimension (using PCA or a learnable embedding layer) to make learning the interaction matrix (B) computationally tractable (Therefore, dimension size K is a hyperparameter).
- **Alternatives (z_j):** The items (e.g., colors) are represented by their standard token embeddings (which might also require a dimensionality reduction) (Therefore, dimension size L is also a hyperparameter).
- **Utility Function:** $u_{ij} = \delta_j x_i B z_j^\top + \epsilon_{ij}$
  - δ_j is the "intrinsic" preference (context-free).
  - B is the interaction matrix that captures how the hidden state modifies preferences.

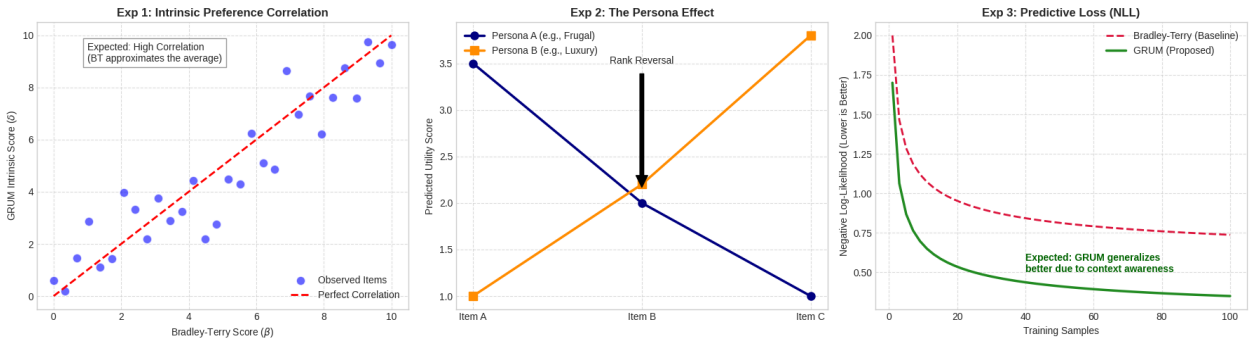

Figure 2: **(Left)** Correlation of intrinsic preferences between GRUM and BT Baseline. **(Center)** Rank reversals illustrating the dependency of preferences on the model's hidden state. **(Right)** Superior predictive performance (lower NLL) of GRUM compared to Bradley-Terry.

2.2 Experiment 1: Intrinsic Preference Validation

We will train both a baseline BT model and our GRUM on the same comparison data from the Qwen-2.5 model family, and compare the learned parameters.

Conjecture: The GRUM's intrinsic parameters (δ) will highly correlate with the BT parameters (β). We expect BT to act as the "average" of the detailed preferences captured by GRUM. See Fig. 2 Left.

2.3 Experiment 2: The persona Effect

We will analyze the interaction matrix B by predicting preferences under two distinct hidden states (e.g., one generated by a "Frugal" prompt vs. a "Luxury" prompt).

Conjecture: The matrix B will be non-zero, causing "Rank Reversals". For example, the Frugal state might rank "Item A" high, while the Luxury state ranks it low, proving preferences depend on the hidden state. See Fig. 2 Center.

To ensure the identified personas represent stable internal states, rather than sensitivity to specific phrasing, we will measure the similarity within a persona cluster and dissimilarity between persona clusters (i.e., each prompt induces an agent's persona x_i, we will look at similarity between the embeddings of prompts with similar semantic meanings such as "You prefer budget options" and "Act as a cost-conscious buyer" and also measure dissimilarity between "Frugal persona" prompts and "Luxury persona" prompts

2.4 Experiment 3: Predictive Performance

We will compare the Negative Log-Likelihood (NLL) of both models on a held-out test set of new prompts.

Conjecture: GRUM will achieve a lower NLL (better fit) than BT. Since GRUM is condition on the hidden state, it can predict the specific variation in preference, whereas BT has to treat that variation as random error. See Fig. 2 Right.

References

Berry, Steven T et al. (1993). *Automobile prices in market equilibrium: Part I and II.*

Chu, Wei and Zoubin Ghahramani (2005). "Preference learning with Gaussian processes". In: *Proceedings of the 22nd international conference on Machine learning*, pp. 137–144.

Daee, Pedram et al. (2017). "Knowledge elicitation via sequential probabilistic inference for high-dimensional prediction". In: *Machine Learning* 106.9, pp. 1599–1620.

Ford Jr, Lester R (1957). "Solution of a ranking problem from binary comparisons". In: *The American Mathematical Monthly* 64.8P2, pp. 28–33.

Soufiani, Hossein Azari (2014). "Revisiting random utility models". PhD thesis. Harvard University.